

Introduction to Tensor Analysis and CMS PS Ch 1-2

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1 Why Tensor Calculus?

1.1 Exercise 1.

Suppose that the temperature field T is given by the function $F(x, y) = x^2 e^y$ in coordinates x, y . Determine the function $F(x', y')$, which gives the temperature field T in coordinates x', y' .

sol.

$$F'(x', y') = F(2x', 2y') = F(x, y) = (2x')^2 e^{2y'} \quad \blacksquare$$

1.2 Exercise 2.

Confirm the equation (1.5) for the functions $F(x, y)$ and $F'(x', y')$ derived in the preceding exercise.

sol.

From here, the equation is:

$$\frac{\partial F'(x', y')}{\partial x'} = 2 \frac{\partial F(x, y)}{\partial x} \quad (1)$$

Since

$$\frac{\partial F(x, y)}{\partial x} = 2xe^y, \quad \frac{\partial F'}{\partial x'} = \frac{\partial F'}{\partial x} \cdot \frac{\partial x}{\partial x'} = (2xe^y)(2) = 2 \cdot \frac{\partial F}{\partial x} \quad \blacksquare$$

1.3 Exercise 3.

Show that the equation

$$\nabla T = \frac{1}{\sqrt{\mathbf{i} \cdot \mathbf{i}}} \frac{\partial F}{\partial x} + \frac{1}{\sqrt{\mathbf{j} \cdot \mathbf{j}}} \frac{\partial F}{\partial y} \quad (2)$$

yields the same result for all rescalings of Cartesian coordinates.

sol.

Since $\mathbf{i}' = 2\mathbf{i}$ and $\mathbf{j}' = 2\mathbf{j}$, $\sqrt{\mathbf{i}' \cdot \mathbf{i}'} = \sqrt{2^2 \cdot \mathbf{i} \cdot \mathbf{i}} = 2$ and same as $\mathbf{j}$.

For the partial derivatives, we've already calculated it from above. Thus, the expression for transformed coordinate is:

$$\nabla T' = \frac{1}{\sqrt{\mathbf{i}' \cdot \mathbf{i}'}} \frac{\partial F'}{\partial x'} + \frac{1}{\sqrt{\mathbf{j}' \cdot \mathbf{j}'}} \frac{\partial F'}{\partial y'} = \frac{1}{2 \cdot \sqrt{\mathbf{i} \cdot \mathbf{i}}} (2 \cdot \frac{\partial F}{\partial x}) + \frac{1}{2 \cdot \sqrt{\mathbf{j} \cdot \mathbf{j}}} (2 \cdot \frac{\partial F}{\partial y}) = \frac{1}{\sqrt{\mathbf{i} \cdot \mathbf{i}}} \frac{\partial F}{\partial x} + \frac{1}{\sqrt{\mathbf{j} \cdot \mathbf{j}}} \frac{\partial F}{\partial y} = \nabla T$$

thus

$$\nabla T' = \nabla T$$

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1.4 Exercise 4.

Show that equation (1.2) yields the same expression in all Cartesian coordinates. The key to this exercise is the fact that any two Cartesian coordinate systems x, y and x', y' are related by the equation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \quad (3)$$

What is the physical interpretation of the numbers a, b , and α ?

sol.

Consider the mapping:

$$\begin{array}{ccc} X \in \mathbb{R}^2 & \xrightarrow{P(X)} & X' \in \mathbb{R}^2 \\ & \searrow F(X) & \swarrow F'(X') \\ & T \in \mathbb{R} & \end{array}$$

For the mapping $P : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, the Jacobian matrix of the function is

$$DP(X) = \begin{bmatrix} \frac{\partial x'}{\partial x} & \frac{\partial x'}{\partial y} \\ \frac{\partial y'}{\partial x} & \frac{\partial y'}{\partial y} \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

thus, the inverse of the Jacobian matrix is

$$DP^{-1}(X') = (DP(X'))^{-1} = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$

From the mapping $F' : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \rightarrow \mathbb{R}$, $F'(X')$ can be calculated by the chain rule:

$$\begin{aligned} DF'(X') &= D(F \circ P^{-1})(X') = DF(X)DP^{-1}(X') = \begin{bmatrix} \frac{\partial F}{\partial x} & \frac{\partial F}{\partial y} \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \\ &\longrightarrow \begin{bmatrix} \frac{\partial F'}{\partial x'} & \frac{\partial F'}{\partial y'} \end{bmatrix} = \begin{bmatrix} \frac{\partial F}{\partial x} & \frac{\partial F}{\partial y} \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \end{aligned}$$

Now, the equation (1.2) can be rewritten as:

$$\nabla T' = \frac{\partial F'}{\partial x'} \mathbf{i}' + \frac{\partial F'}{\partial y'} \mathbf{j}' = \begin{bmatrix} \frac{\partial F'}{\partial x'} & \frac{\partial F'}{\partial y'} \end{bmatrix} \begin{bmatrix} \mathbf{i}' \\ \mathbf{j}' \end{bmatrix} = \begin{bmatrix} \frac{\partial F'}{\partial x'} & \frac{\partial F'}{\partial y'} \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \mathbf{i} \\ \mathbf{j} \end{bmatrix}$$

Thus,

$$\nabla T' = \begin{bmatrix} \frac{\partial F'}{\partial x'} & \frac{\partial F'}{\partial y'} \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \mathbf{i} \\ \mathbf{j} \end{bmatrix} = \begin{bmatrix} \frac{\partial F'}{\partial x'} & \frac{\partial F'}{\partial y'} \end{bmatrix} \mathbf{I} \begin{bmatrix} \mathbf{i} \\ \mathbf{j} \end{bmatrix} = \frac{\partial F}{\partial x} \mathbf{i} + \frac{\partial F}{\partial y} \mathbf{j}$$

$$\nabla T' = \nabla T$$



My question: How to determine the basis of an affine transform by algebra?(Of course, it's obvious for geometry.)

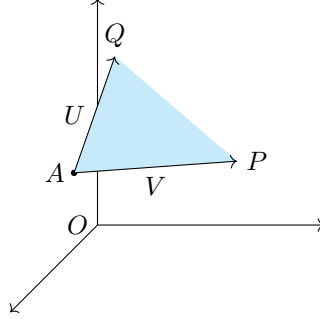
1.5 Exercise 5.

Conclude that equation (1.6(2)) is valid for all orthogonal affine coordinate systems (Euclidean Transform.) Affine coordinates are those with straight coordinate lines.

2 Rules of the Game

Before we begin, consider some geometrical logic that can be applied to n-dimensional space. For the Pythagorean Theorem and law of cosines, we can treat them also in n-dimensional Euclidean space without proof; they are related to only two vectors.

In Linear Algebra, treating two vectors means that there is a subspace spanned by the vectors, and the space itself has a degree of freedom, 2 (already proved from two-dimensional Euclidean Space.) That is, no matter what basis vectors are set and how many vectors are in the space, we only cover two-dimensional space such that we can still use the properties:



Law of Cosines

$$\|U - V\|^2 = \|U\|^2 + \|V\|^2 - 2\|U\|\|V\|\cos\alpha \quad (4)$$

$$\|U - V\|^2 = \|U\|^2 + \|V\|^2 - 2U \cdot V \quad (5)$$

Pythagorean Theorem, in case of $\alpha = 0^\circ$

$$\|U - V\|^2 = \|U\|^2 + \|V\|^2 \quad (6)$$

From the Law of Cosines, we can also find the formula for the dot product without the angle:

$$U \cdot V = \frac{\|U\|^2 + \|V\|^2 - \|U - V\|^2}{2}$$

$$U \cdot (-V) = \frac{\|U\|^2 + \|V\|^2 - \|U + V\|^2}{2}$$

which gives the formula

$$U \cdot V = \frac{\|U + V\|^2 - \|U - V\|^2}{4} \quad (7)$$

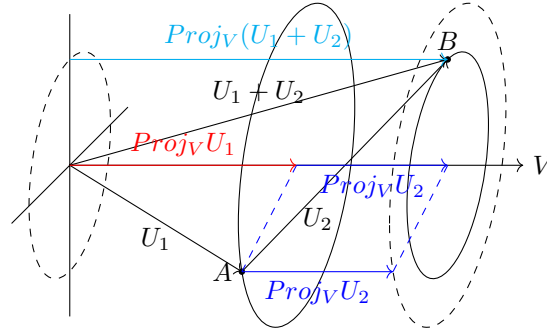
2.1 Exercise 6.

To demonstrate geometrically that the dot product is distributive, we can use the properties of vectors and the geometric interpretation of the dot product.

sol.

Consider three vectors U_1 , U_2 and V . Since the dimension of a subspace that can be made by them is at most three, they can be treated as vectors in three-dimensional space by a well-defined basis.

Now, they are represented in a cylinder parallel to the direction of V :



From the figure, Projection of $U_1 + U_2$ now has same direction with $Proj_V U_1$ and $Proj_V U_2$ such that the following holds:

$$Proj_V(U_1+U_2) = Proj_V U_1 + Proj_V U_2 \longleftrightarrow \|V\| (Proj_V(U_1+U_2)) = \|V\| (Proj_V U_1 + Proj_V U_2)$$

$$\longrightarrow (U_1 + U_2) \cdot V = U_1 \cdot V + U_2 \cdot V$$

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References

- [1] Pavel Grinfeld. *Introduction to Tensor Analysis and the Calculus of Moving Surfaces*. Springer, 2013.
- [2] StackExchange. *Proving that the dot product is distributive?* 2015. URL: <https://math.stackexchange.com/questions/1109142/proving-that-the-dot-product-is-distributive> (visited on 07/19/2025).